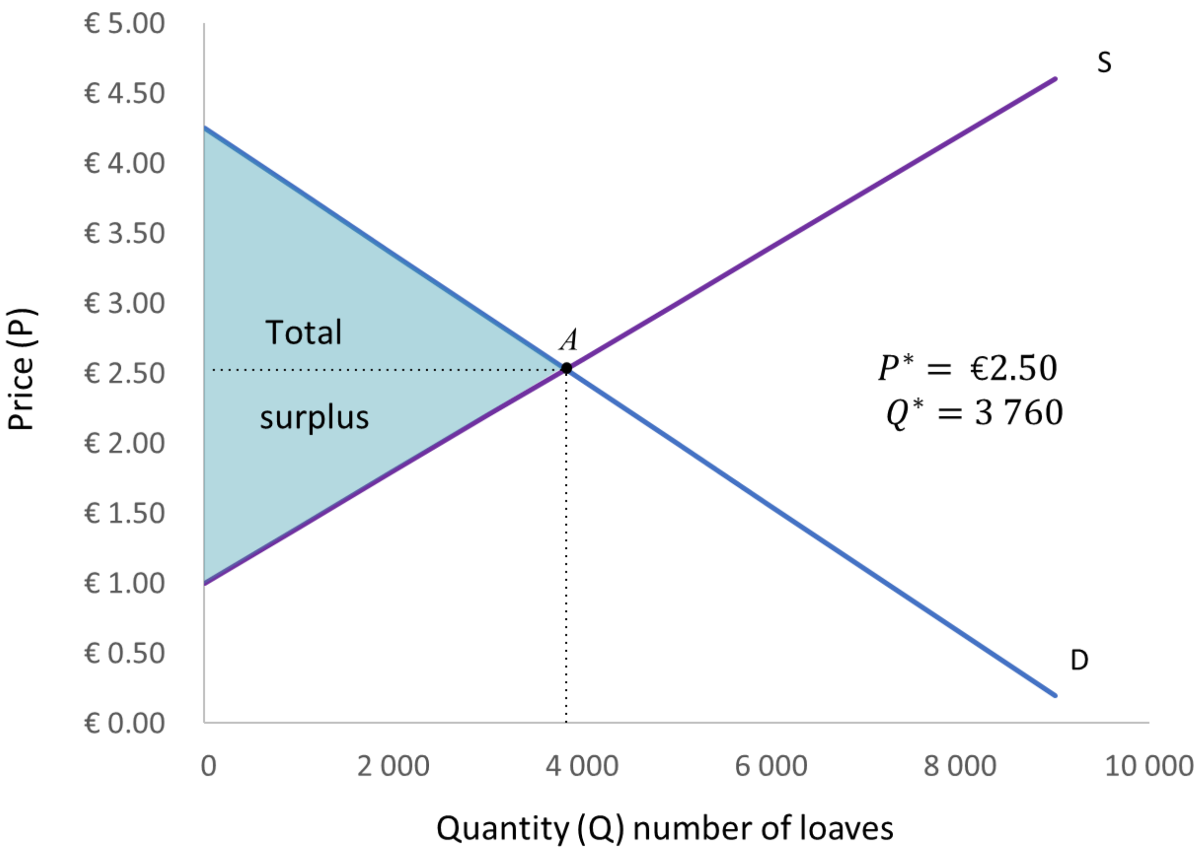
Unit 7&8 Part 5: Taxes and price controls

UNIT 7&8 Part 5 Overview

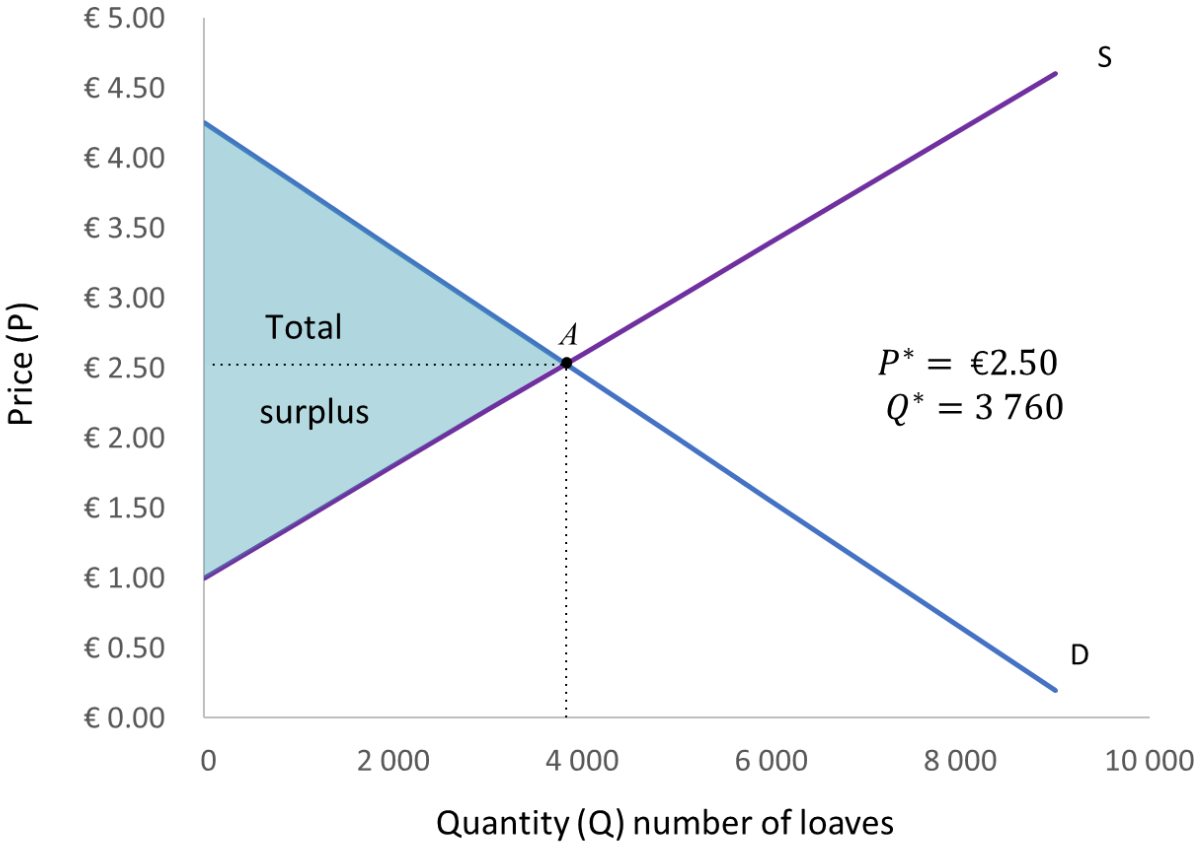
- Government intervention in competitive markets
- Taxes: Price wedges and equilibrium adjustment
- Tax incidence
- Welfare effects of taxes
- Using taxes to change behaviour
- Price rationing
- Price controls in perfectly competitive markets
- Price ceilings: Binding vs non-binding
- Rent ceilings: Shortages, rents and search costs
- Rent ceilings: Welfare effects; efficiency and fairness
- Price floors: Binding vs non-binding
- Minimum wages

Government intervention in competitive markets



- We've seen that when rational buyers and sellers are free to make choices in their own interests in competitive markets, the outcome is Pareto efficient.
- It is impossible to make someone better off without making someone else worse off.
- WTP = WTA → Total surplus is maximised.
- $Q_d = Q_S \rightarrow$ Buyers and sellers can execute their decisions – there are no shortages or surpluses

Government intervention in competitive markets



- Government (or other) interventions that disturb the equilibrium so that the quantity supplied is no longer where WTP = WTA will result in inefficiency.
- E.g. taxes and price controls
- In some cases, this can be justified:
  - if it corrects another inefficiency elsewhere (e.g. public goods)
  - if it results in a fairer outcome

8.12 Taxes

- Governments use taxes to:
  - Finance government spending
  - Redistribute resources
  - Discourage “bad” behavior
- Taxes
  - Increase consumer prices
  - Change buyers’ and sellers’ decisions
  - Can generate deadweight loss
- Taxes *improve* or *reduce* overall welfare depending on how the government spends the revenue.
- Do the taxes you pay increase or decrease your welfare?

Using taxes to raise revenue – salt example

- For most of history, humans have needed salt to survive and to preserve food.
- Rulers historically taxed salt to raise revenue because:
  - demand is relatively inelastic.
  - the tax is relatively easy to administer.
- In ancient Rome, soldiers' salaries were paid in salt.
- Being "worth one's salt" means to be deserving of one's pay.

Salt mounds in Salar de Uyuni, Bolivia

Salt taxes

The equilibrium price in the perfectly competitive market for salt is currently R10 per kilogram. The government decides to levy a tax of R2 per kilogram. What will the new price of salt be?

Not R12!

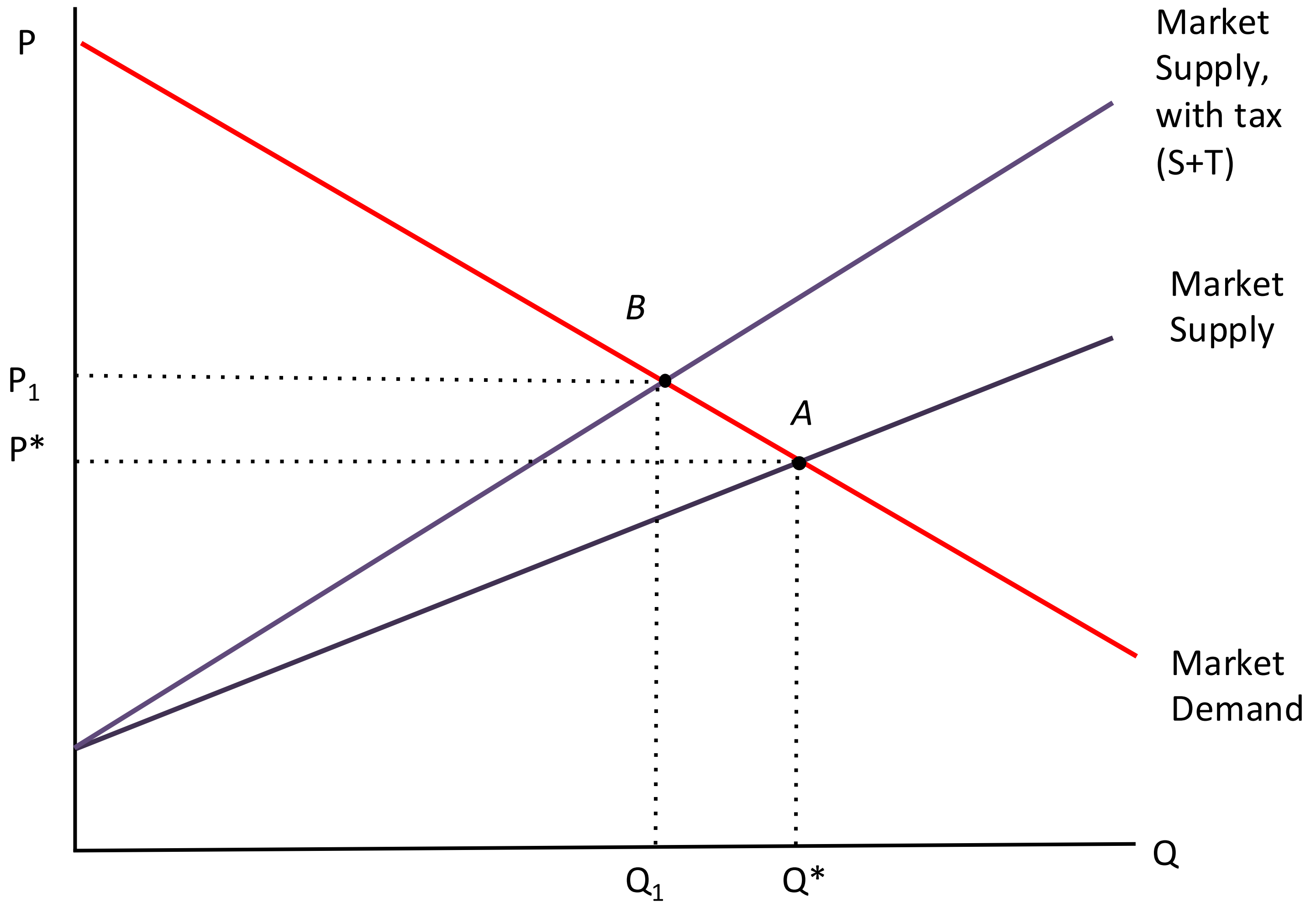
What we know is that the price buyers pay per kilogram will exceed the price sellers get by R2 → a tax creates a **price wedge** equal to the size of the unit tax (tax amount per unit).

But taxes also affect incentives to buy and sell, so the market will have to readjust to a new equilibrium, which affects prices and quantities.

The supply and demand model is useful for analyzing the effects of taxing salt.

Salt taxes

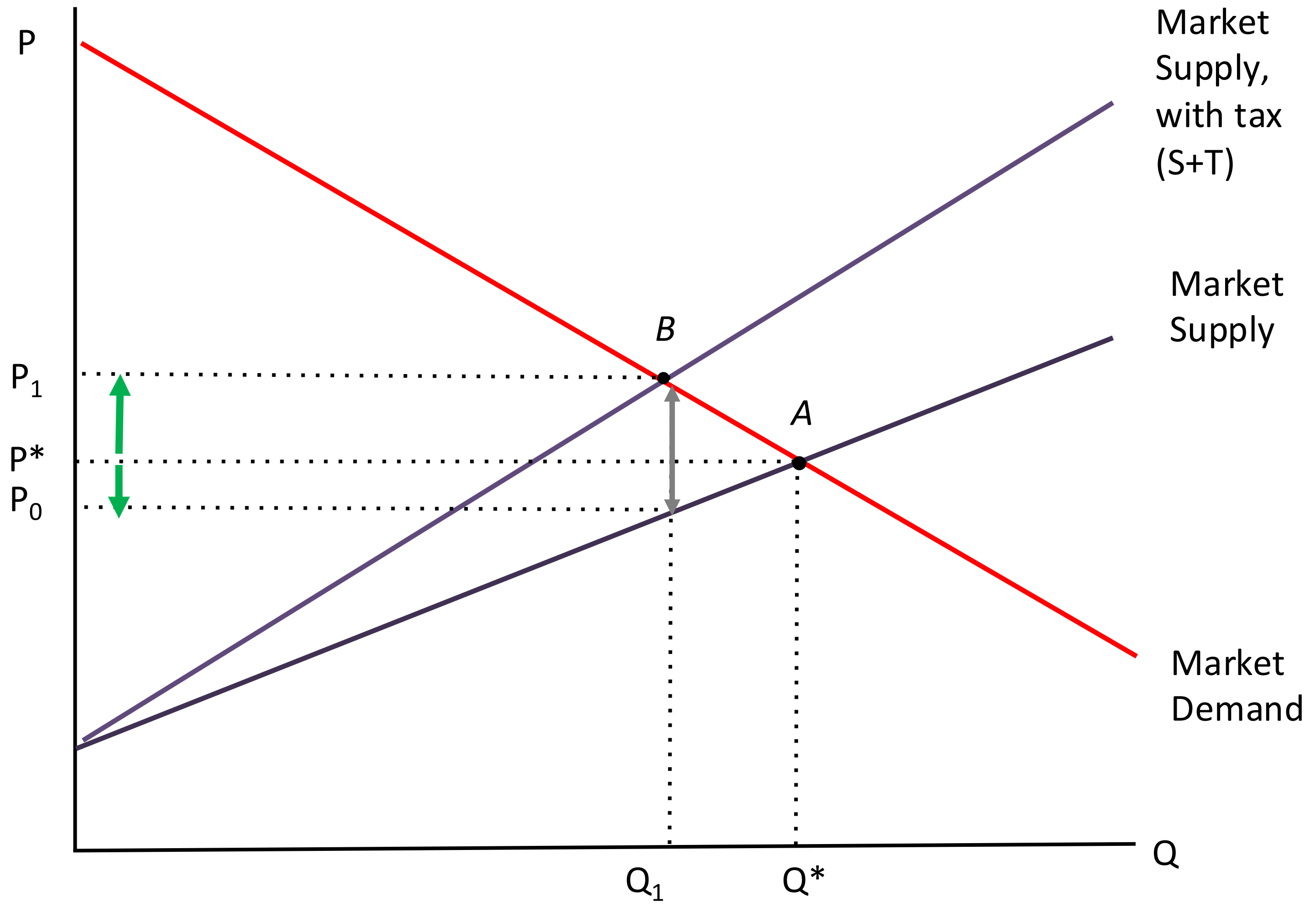
- Initially the market equilibrium is at point A. The price is *P** and the quantity of salt sold is *Q**
- Suppose the government imposes a 30% tax ad valorem on salt:
  - Supply with tax curve = MC*(1 + 30% tax)
  - Changes the market equilibrium from A to B.
  - At P_1 and Q_1



Salt taxes

- The new competitive equilibrium:
  - Increases the price paid by consumers from P^* to P_1 (by less than 30%)
  - Reduces the price received by producers from P^* to P_0
  - The market clears:

    Qs at $P_0 =$ Qd at P_1
  - $P_1 - P_0$ is the tax paid to government per unit of salt sold



Class exercise

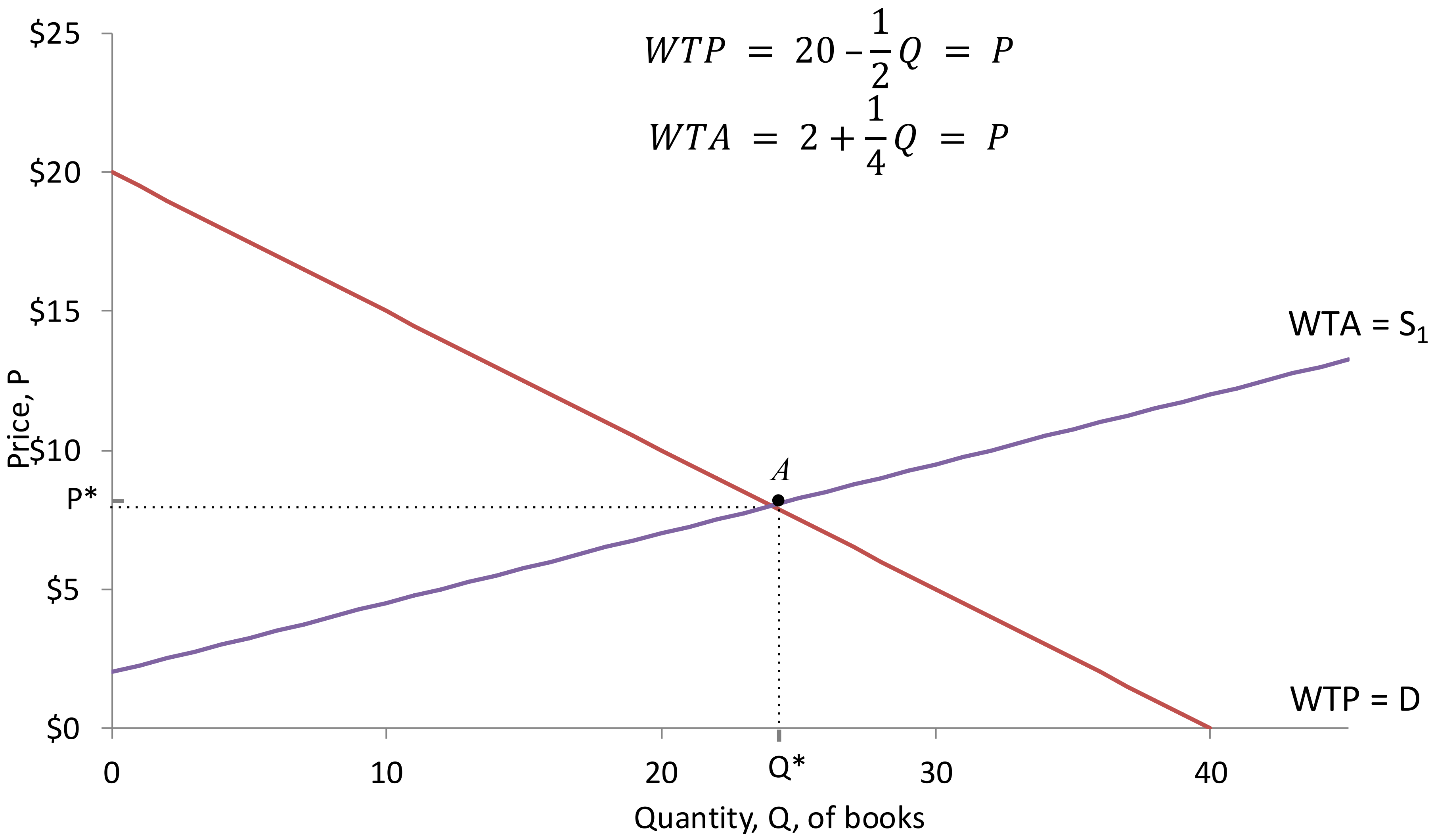
Consider the market for used textbooks. Supply and demand are given by

$$P = 20 - \frac{1}{2}Qd$$

$$P = 2 + \frac{1}{4}Qs$$

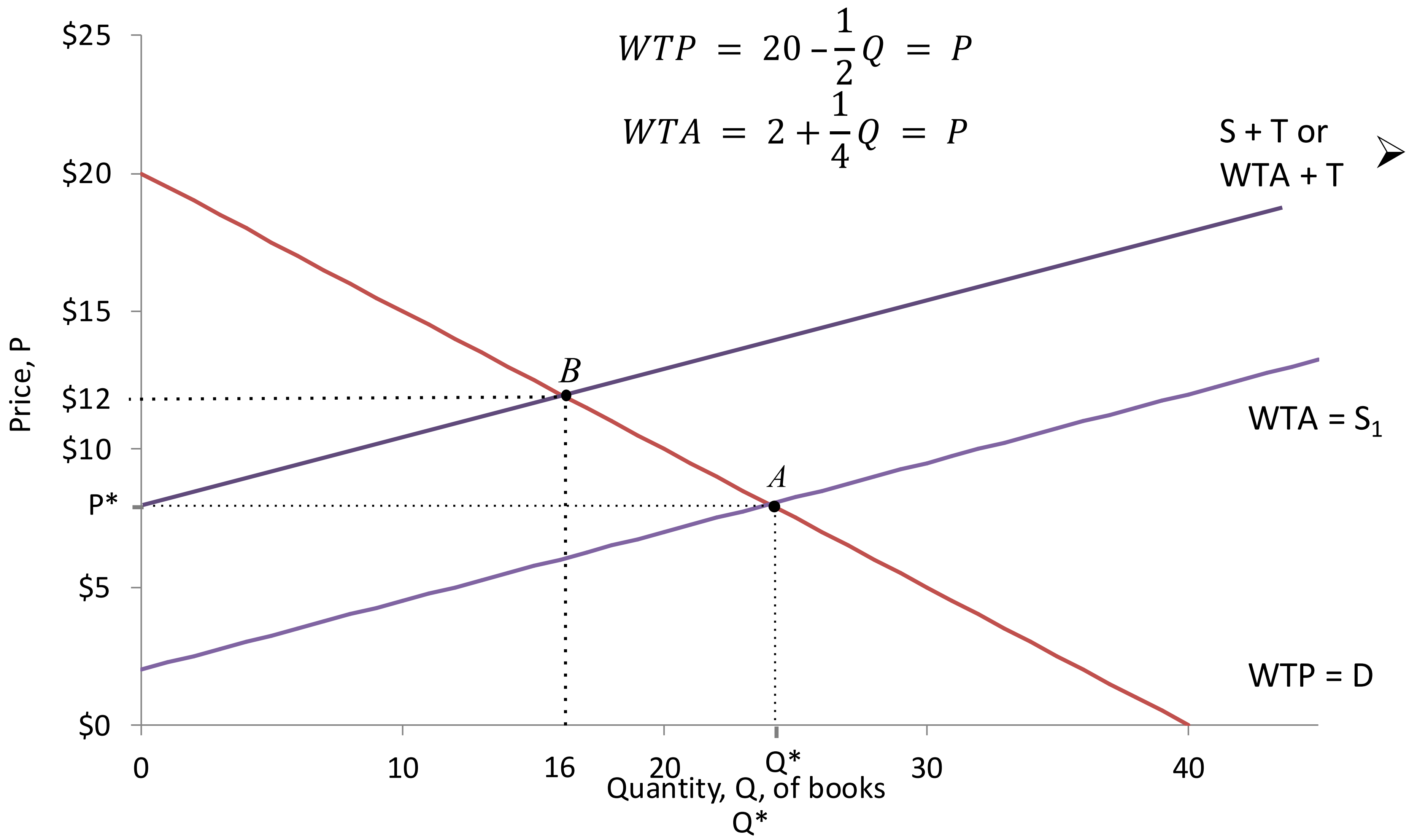
1. What is the market equilibrium P and Q without a tax?
2. What happens to P and Q when the government decides to levy a tax of $6 per used textbook sold?

Answer: Used textbook tax



1. What is the competitive equilibrium?
    - $20 - \frac{1}{2}Q = 2 + \frac{1}{4}Q$
    - $18 = \frac{3}{4}Q$
    - $Q^* = 24$
    - $P^* = \$8$

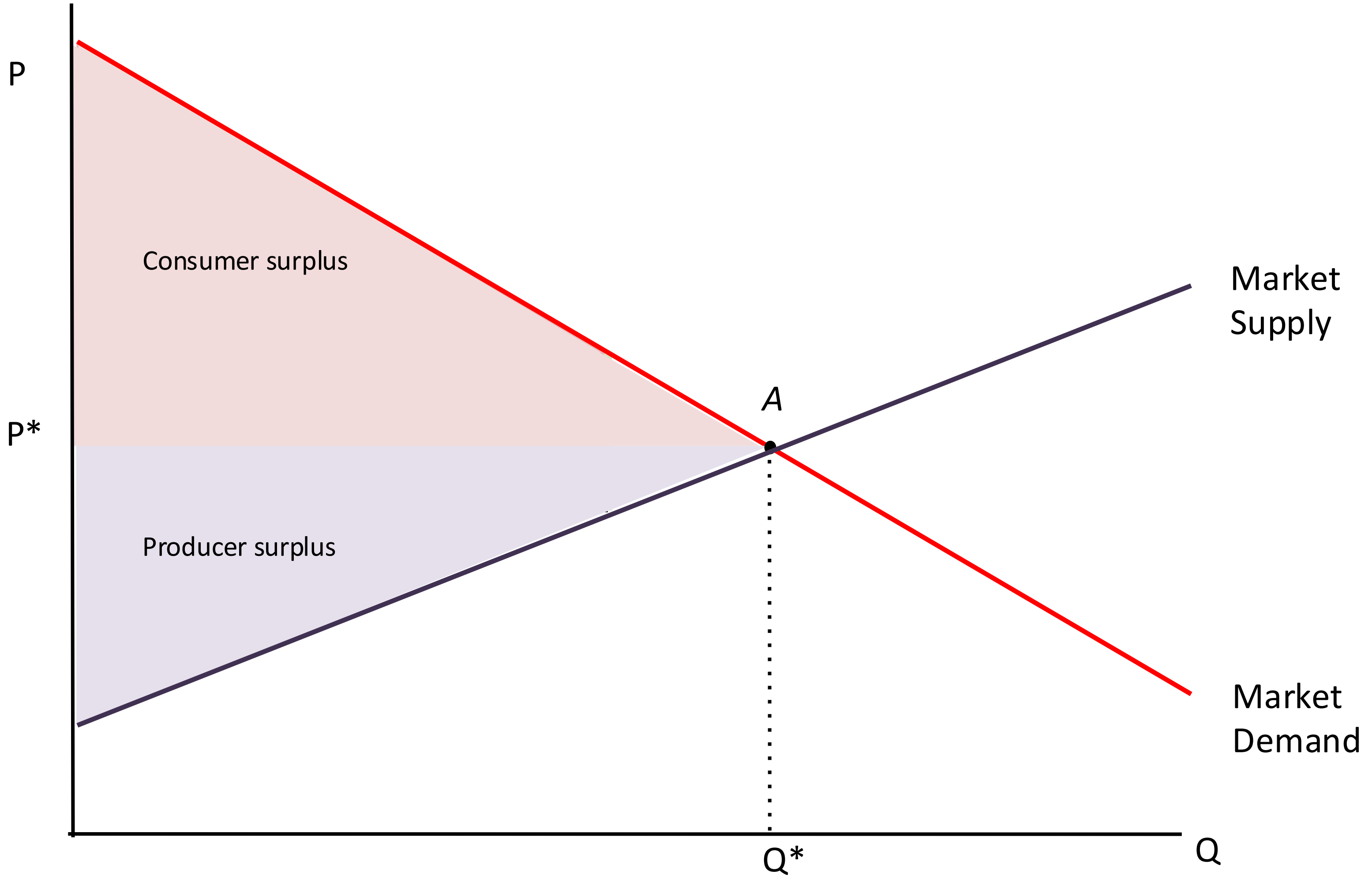
Used textbook tax



- Competitive equilibrium without tax
  - $Q^* = 24, P^* = \$8$
- Suppose the government imposes a $6 tax on used textbooks.
  - $WTA + T = 8 + \frac{1}{4}Q$
  - $WTA + T = WTP$
  - $8 + \frac{1}{4}Q = 20 - \frac{1}{2}Q$
  - $12 = \frac{3}{4}Q$
  - $Q^* = 16\ , P^* = \$12$
  - Seller's price = $P^* - T = \$6$
  - Tax revenue = $\$6 \times 16 = \$96$

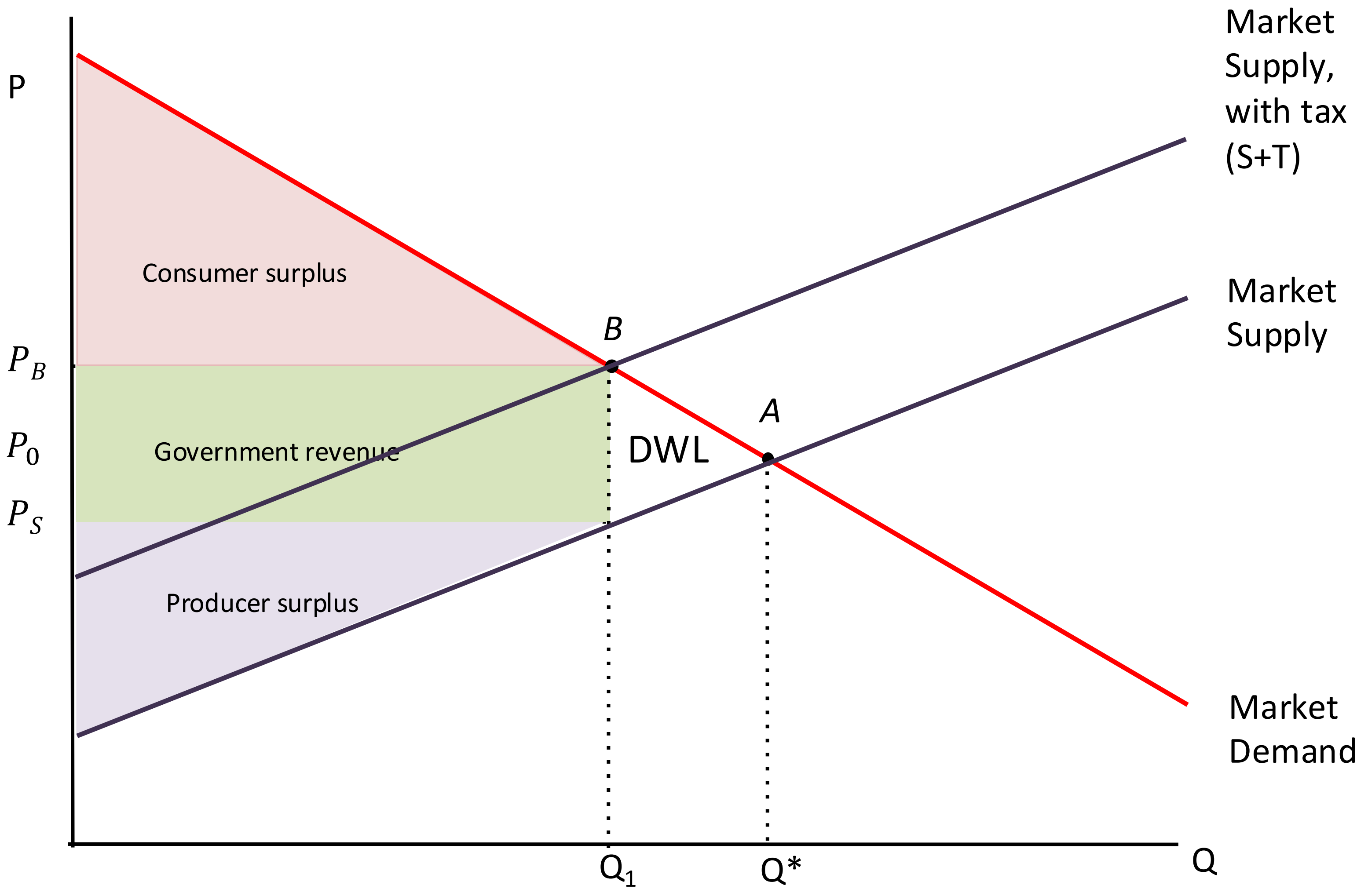
Salt taxes – effect on consumers and producers

- Without a tax, the competitive market maximizes the total surplus.



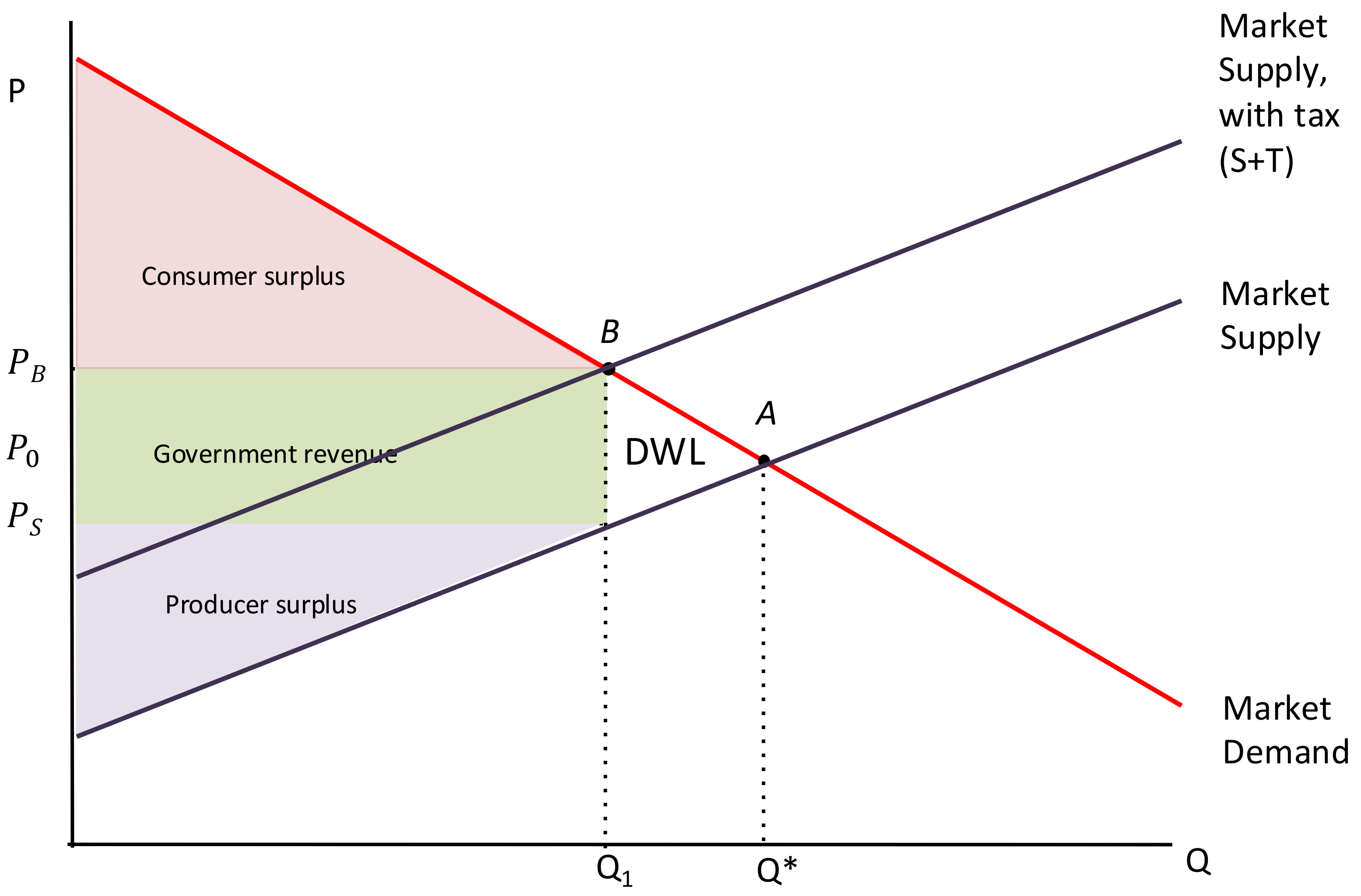
Salt taxes – effect on consumers and producers

- Without a tax, the competitive market maximizes the total surplus.
- The tax reduces both the consumer and producer surplus
- The salt tax:
  - Raises government revenue
  - Reduces the total surplus
  - Creates deadweight loss (DWL)
- Government revenue is not a social loss (assuming it is spent on socially valuable goods and service)

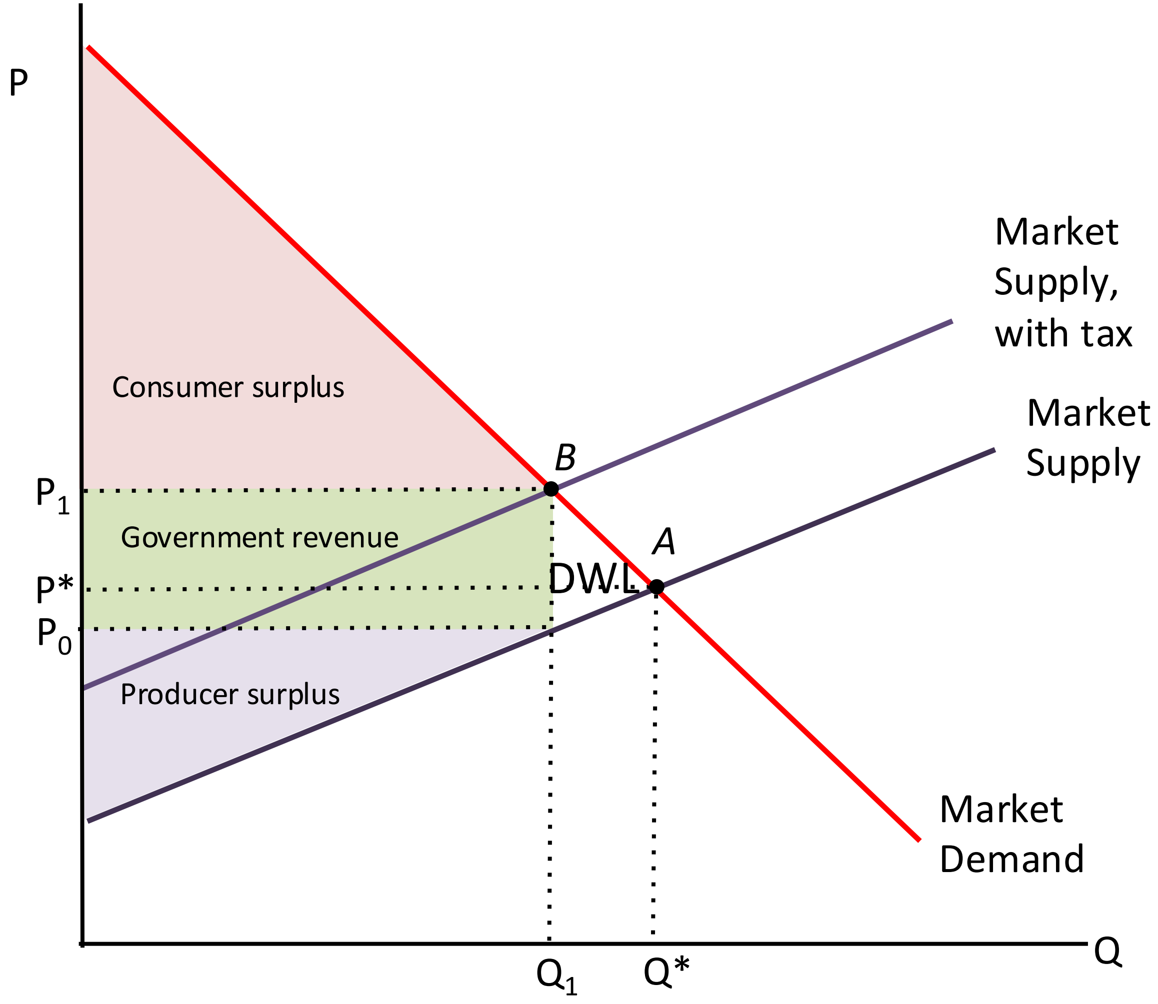


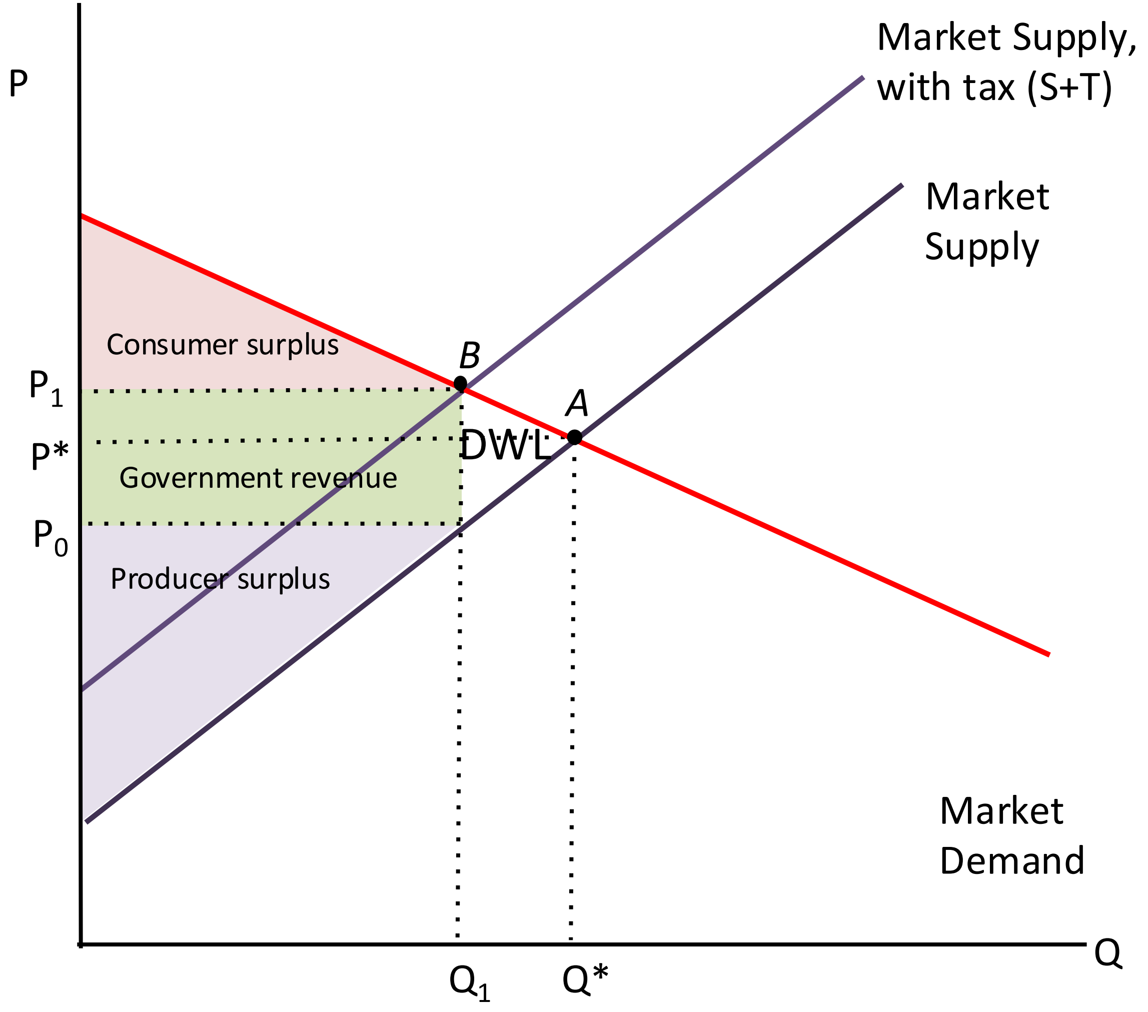
Salt taxes – Tax incidence

- Tax incidence is the division of the tax burden between consumers and producers
  - Consumer's burden $= P_B - P_0$
  - Producer's burden $= P_0 - P_S$
- The tax burden/tax incidence falls mostly on,
  - consumers when the demand curve is more inelastic (steep)
  - producers when the supply curve is more inelastic (steep)
- DWL is large when supply and demand curves are more elastic (flatter) – tax then has a larger effect on quantity



Tax incidence and price elasticity

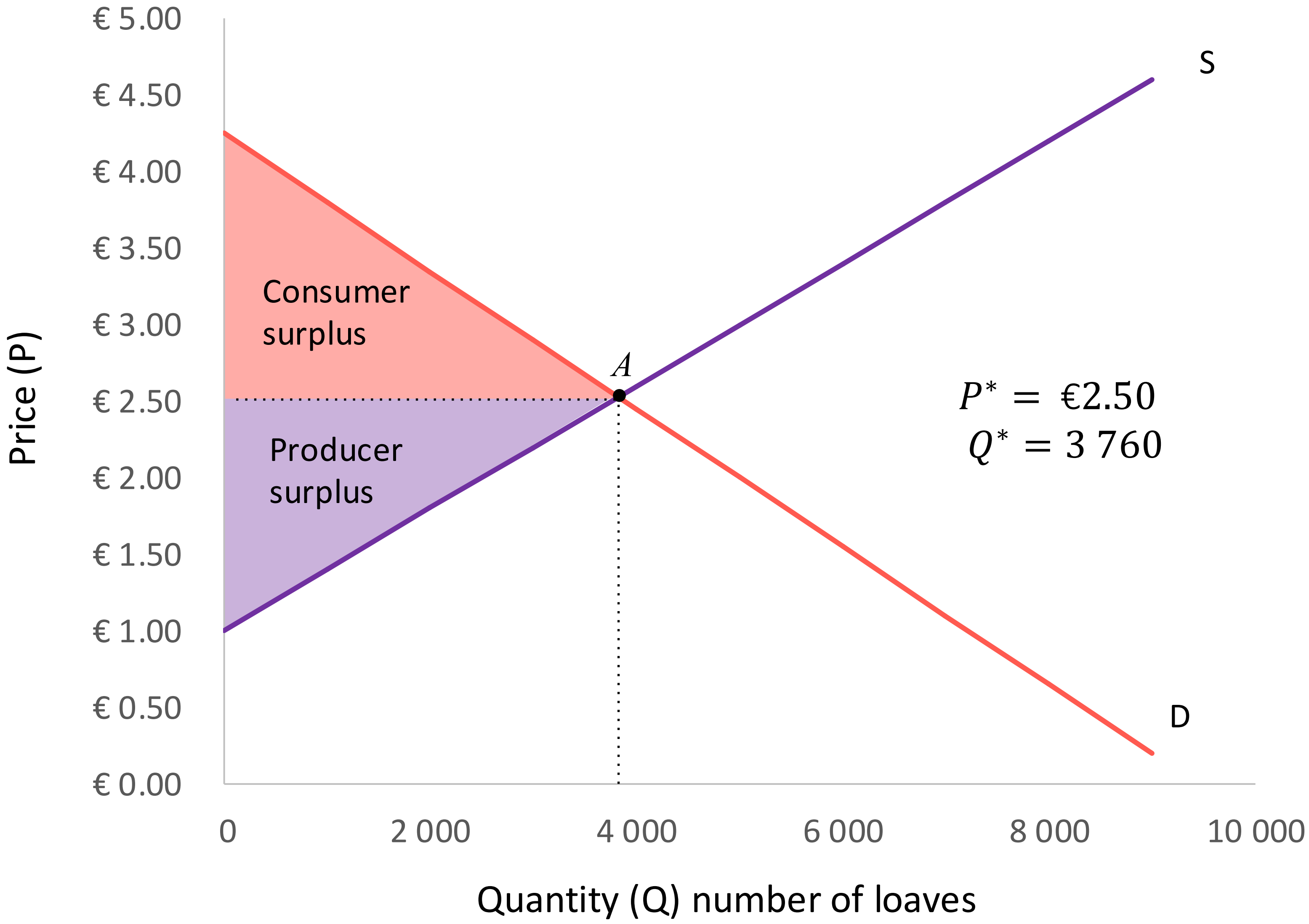




Taxes, welfare and behaviour

- When the demand (or supply) of a good is inelastic, a tax has little impact on the equilibrium quantity and:
  - Raises more revenue
  - Creates less deadweight loss
- The impact of taxes on total surplus, which is a measure of welfare, depends on
  - The size of the deadweight loss
  - How the tax revenue is spent
- Sometimes the point of a tax is not to raise revenue, but to change behaviour.
- Governments see some goods as "bads".
  - Hungary taxes chips
  - US states tax beer and cigarettes
  - Colorado taxes marijuana
  - Sugar Tax in South Africa

Price rationing



In a competitive bread market, the price adjusts until an efficient equilibrium is reached.

But **whom** among the potential consumers actually get the available bread?

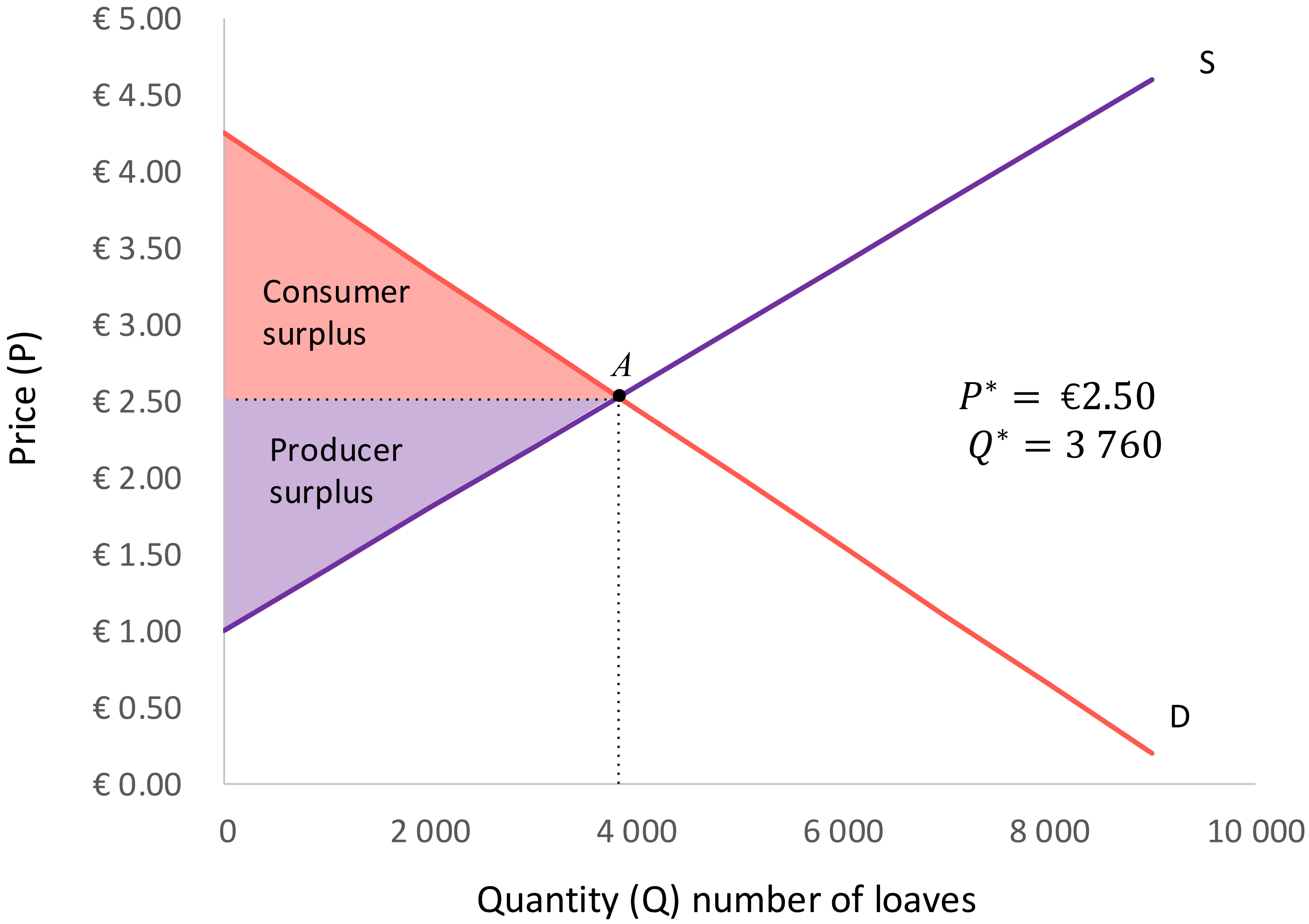
→Those willing (and able) to pay the market price.

→Those not willing (or able) to pay the market price do not get any units.

The price **rations** the available units.

Those who value it the highest (in the WTP sense) get it.

Price rationing



Whom among potential producers actually supply the product?

→Those with WTA ≤ P

→Those with the lowest (marginal) cost

In addition, we know that in the **long run**, competition drives price down to minimum average cost.

Firms with higher cost will face economic losses and exit the market.

Price rationing

- Price adjustments therefore ensure that $Q_s = Q_d$ by rationing the available quantity supplied to consumers with the highest WTP and the available quantity demanded to producers with the lowest WTA.
- What happens when price adjustments are prevented?

Read: Core Unit **8** section **8.12**

8.12 Price controls

- New York yellow taxi cabs operate at a fixed rate (minor adjustments for different times of the day)
- When demand for metered taxis is very high (e.g. during a snowstorm), users can face long waiting times

- With Qd > Qs, the price cannot change to “clear the market”
  - The available taxis are rationed to those who are willing to wait long and those who are lucky
- Uber’s prices, on the other hand, can change substantially based on demand and supply
  - On a particularly stormy evening in December 2013, Uber’s surge-pricing algorithm meant that fares were more than seven times the standard rate
  - The surge in price can lower Qd and increase Qs until a market-clearing price is reached
  - The available taxis are rationed by price to those with the highest WTP

8.12 Price controls

Price regulation by authorities are intended to protect groups of individuals who are regarded as vulnerable or politically sensitive

Examples:

- Rent control: when housing / flat rental prices go up in a city, e.g. due to gentrification, fast economic growth or demographic factors, long time, lower-income renters may no longer be able to afford to live in their houses or neighborhoods
- Prices of food and other basic needs are controlled to make basic needs affordable to the poor
- Fuel/energy prices are kept low to promote economic growth or controlled to prevent price instability
- Minimum wages are intended to ensure that the lowest-paid workers earn a living wage. Centralised bargaining by labour unions in an industry can have similar effects, as wages tend to be driven by large firms and their workers, so the agreed wage functions like a minimum wage on smaller firms.
- Sometimes, price controls are implemented in an attempt to counteract high inflation caused by poor fiscal / monetary policies (e.g. Zimbabwe, Venezuela)

8.12 Price controls

We will consider price controls in the case of imperfect competition in Part 6.

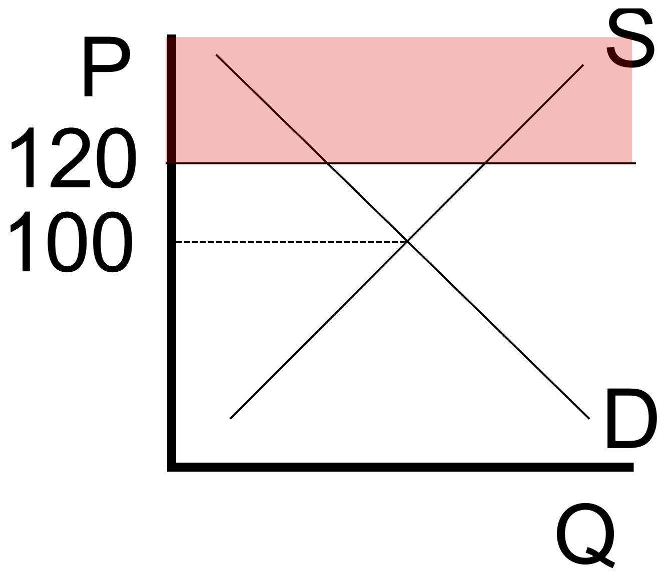
Here, we will consider the case of price controls in **perfectly competitive** markets

- housing markets → Rent controls (price ceiling)
- labour markets → Minimum wage (price floor)

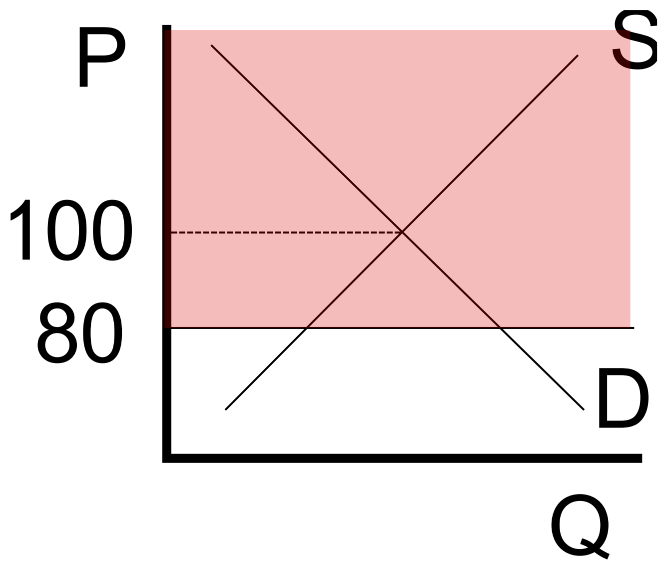
Binding and non-binding price ceilings

- If the equilibrium price with no government intervention is R100, then a price ceiling of R120 would be **non-binding** and have no effect.

  (the shaded area shows illegal prices)

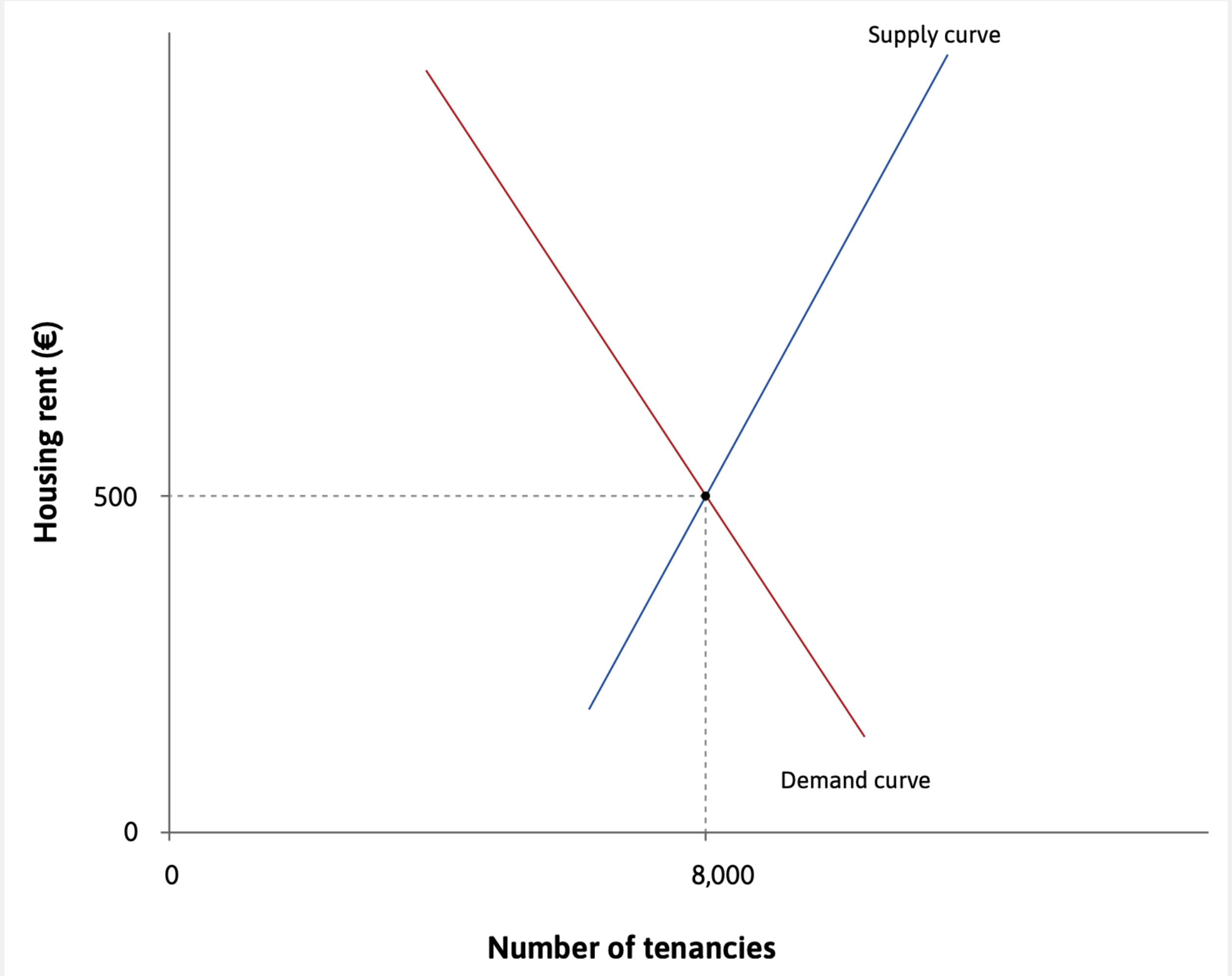


- If the equilibrium price with no government intervention would be R100, then a price ceiling of R80 would be **binding**.



8.12 Rent ceiling

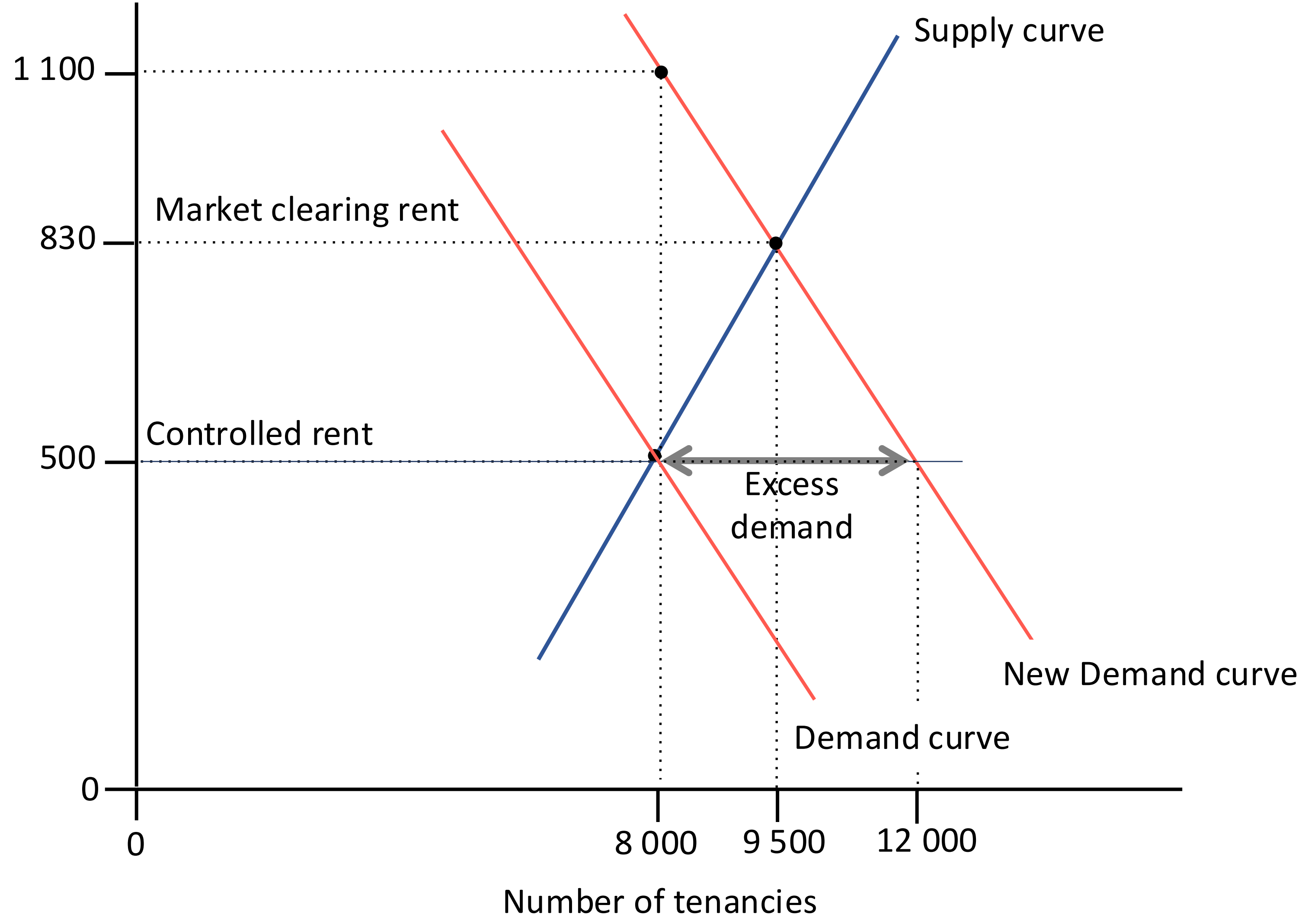
- The market-clearing (Qd = Qs = 8 000) rental price is €500
- Now suppose the demand for rental housing increases



8.12 Rent ceiling

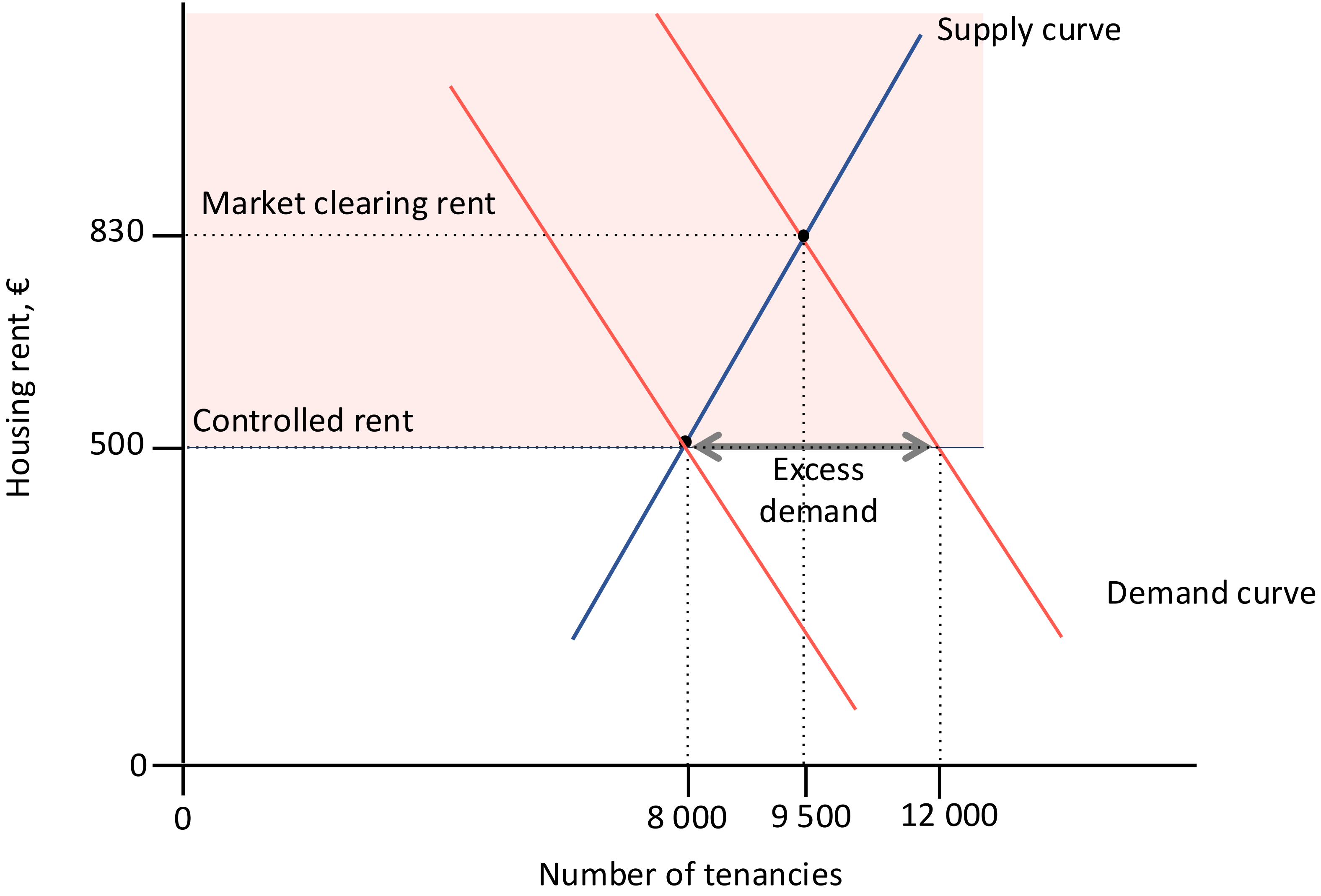
- The market-clearing (Qd = Qs = 8 000) rental price is €500
- Now suppose the demand for rental housing increases

- The market-clearing (Qd = Qs = 9500) rental price is €830



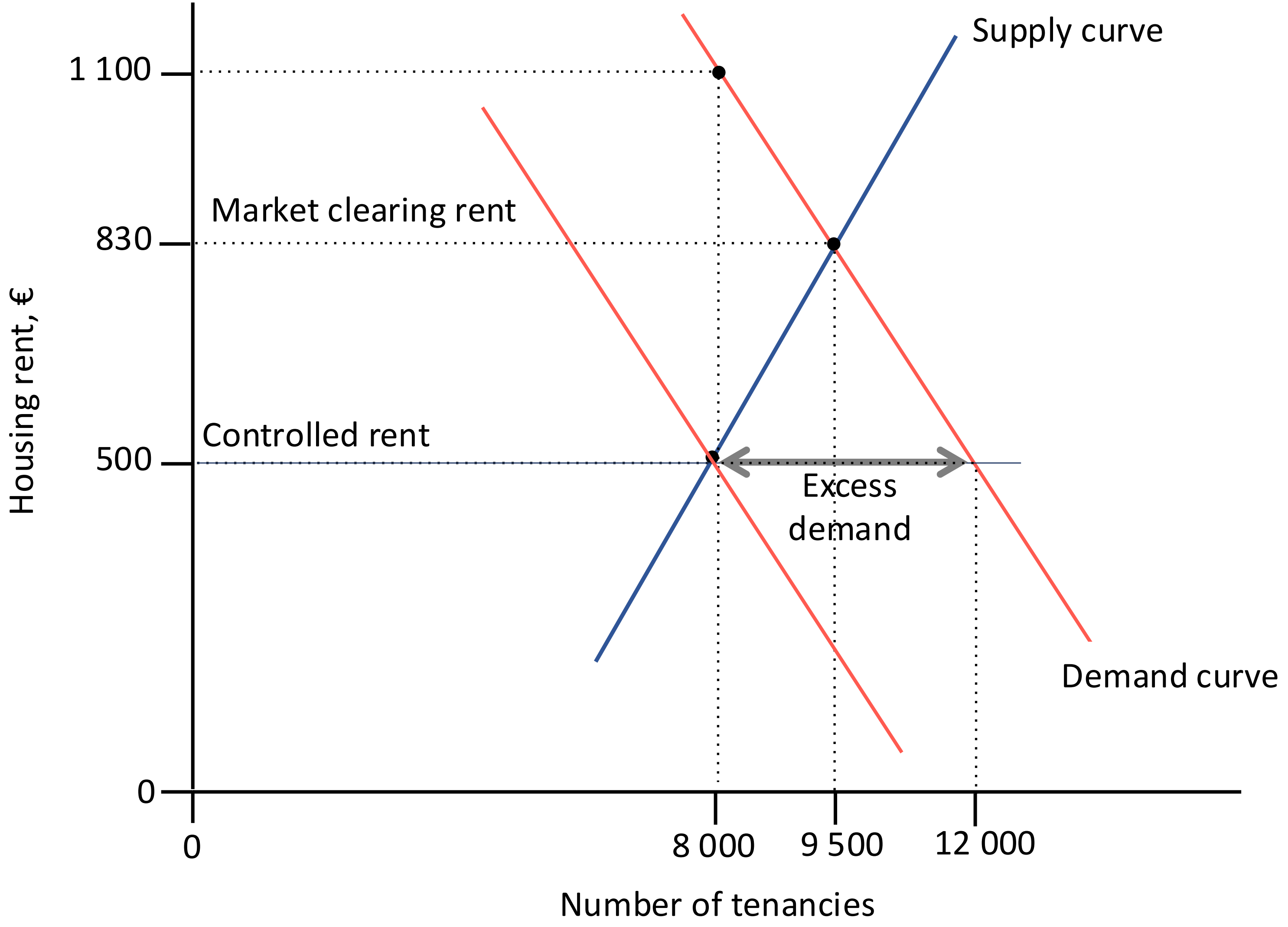
8.12 Rent ceiling

- City authorities may be concerned that the market clearing rent ($830) may be unaffordable to some people
- They impose a rent ceiling of €500
  - The market-clearing rent of €830 is now illegal
- At a controlled rent of €500, Qs = 8 000 and Qd = 12 000
- Only 8 000 units get rented out
- There is a shortage (excess demand) of 4 000 tenancies



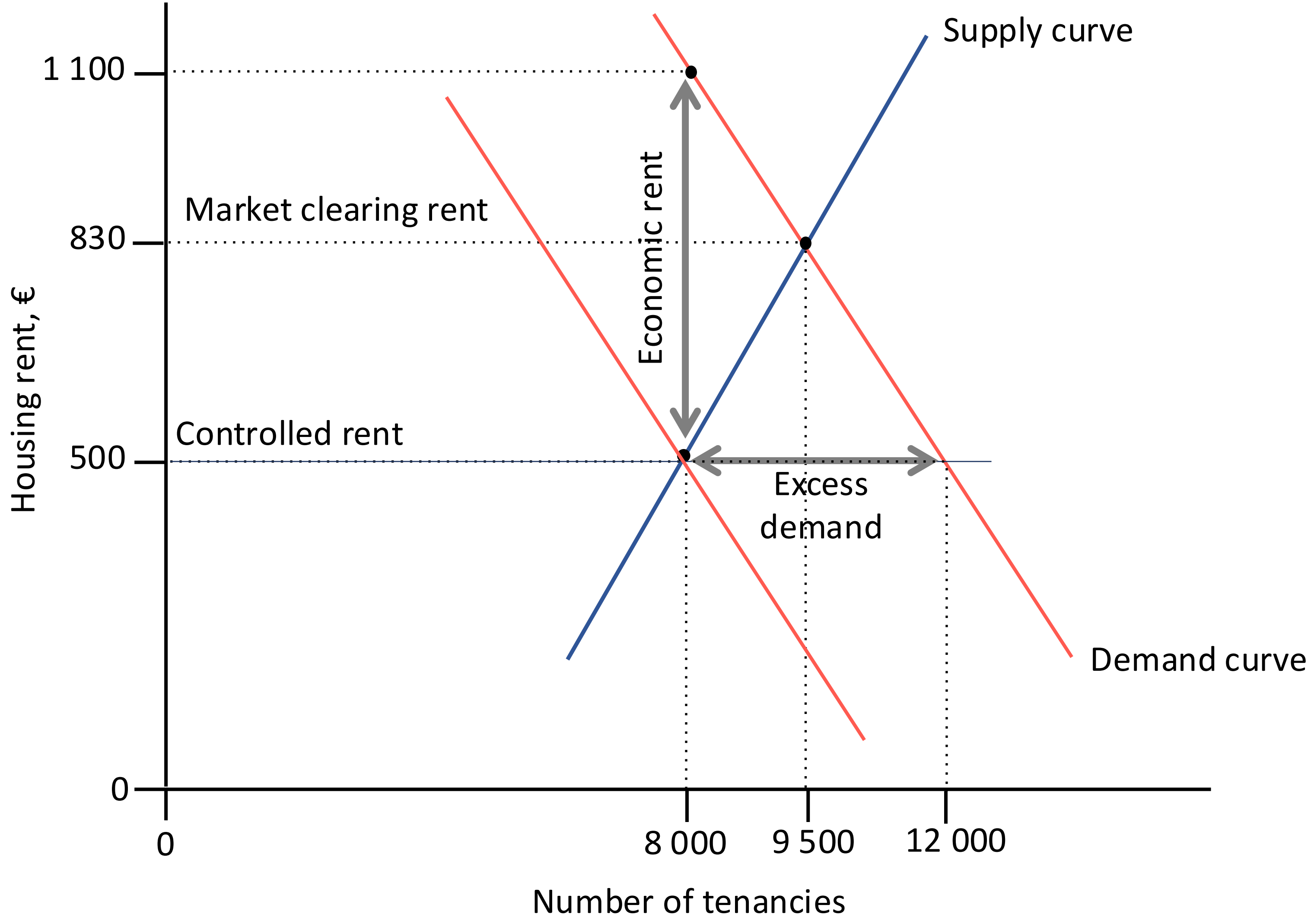
8.12 Rent ceiling

- Housing will not be allocated based on willingness to pay, but rationed in some other way, e.g. first-come-first-serve, luck, willingness to wait/search, social contact networks, discrimination
- This means that even tenants not willing to pay €830 but able to pay the controlled rent of €500 (those intended to be protected) are not guaranteed to get housing



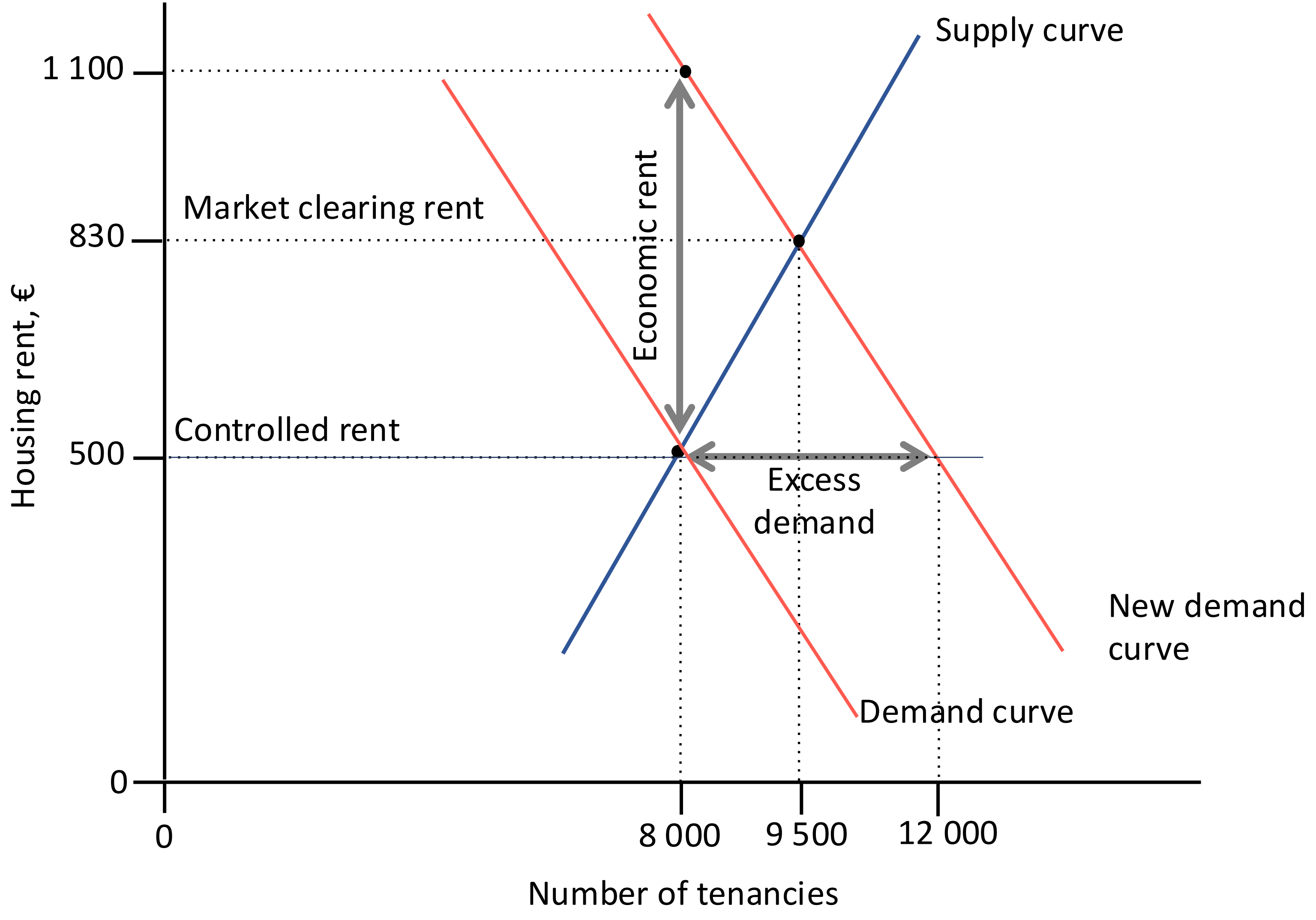
8.12 Rent ceiling

- Notice that 8 000 of the 12 000 potential tenants are willing to pay in excess of €1 100
- The scarcity of rental housing gives rise to potential economic rent
  - Tenants receive economic rent of €600 at the margin



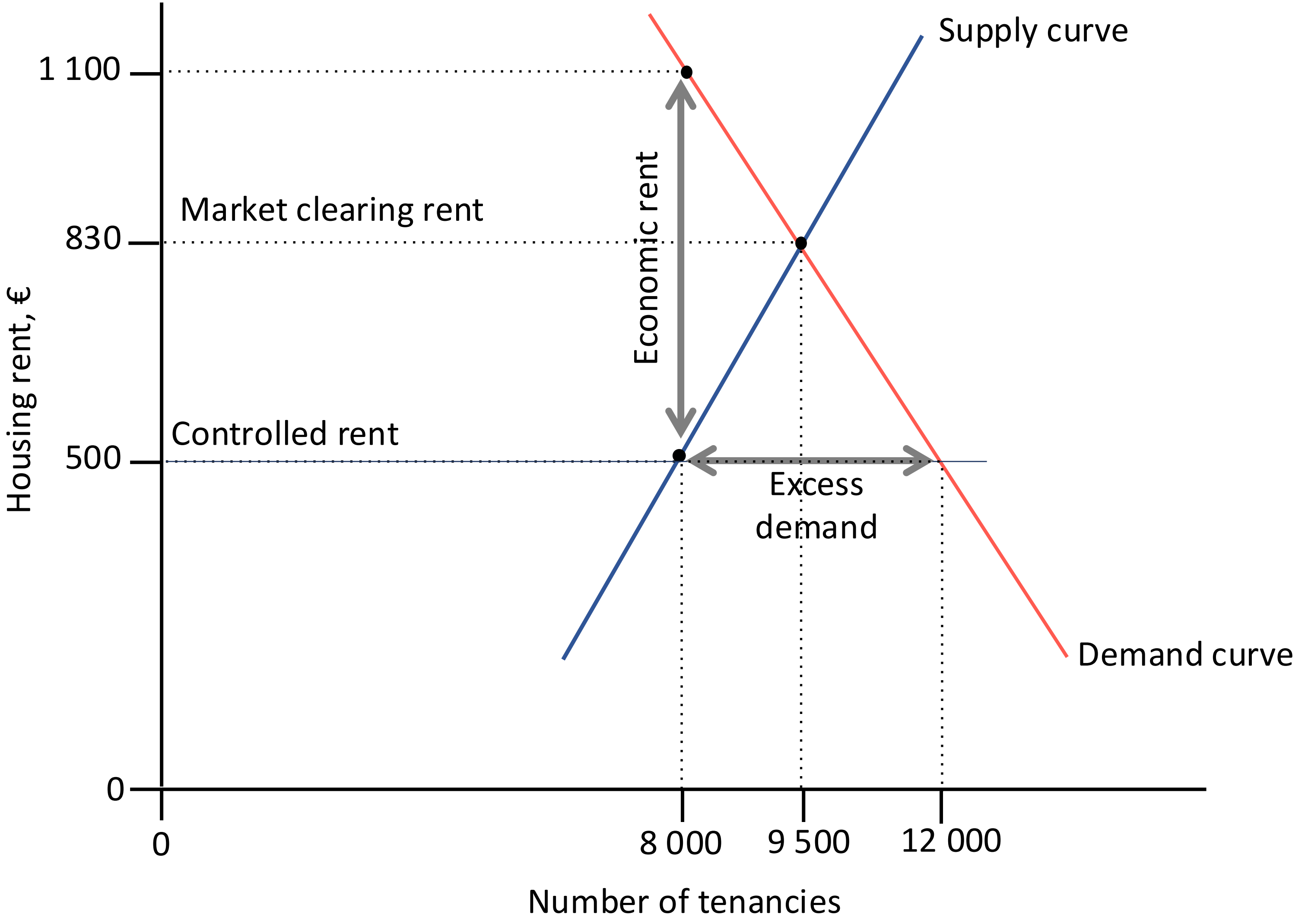
8.12 Rent ceiling

- Some tenants can monetize their rent by subletting (possibly illegal)
- Given the housing shortage and the rent incentive, potential tenants will be willing to expend **search costs**.
    - Willing to pay up to €600 (at the margin) or even more (for those with higher WTP) in order to be lucky enough to get a rent-controlled housing unit
    - Unnecessary search costs represent wastage of scarce resources to society



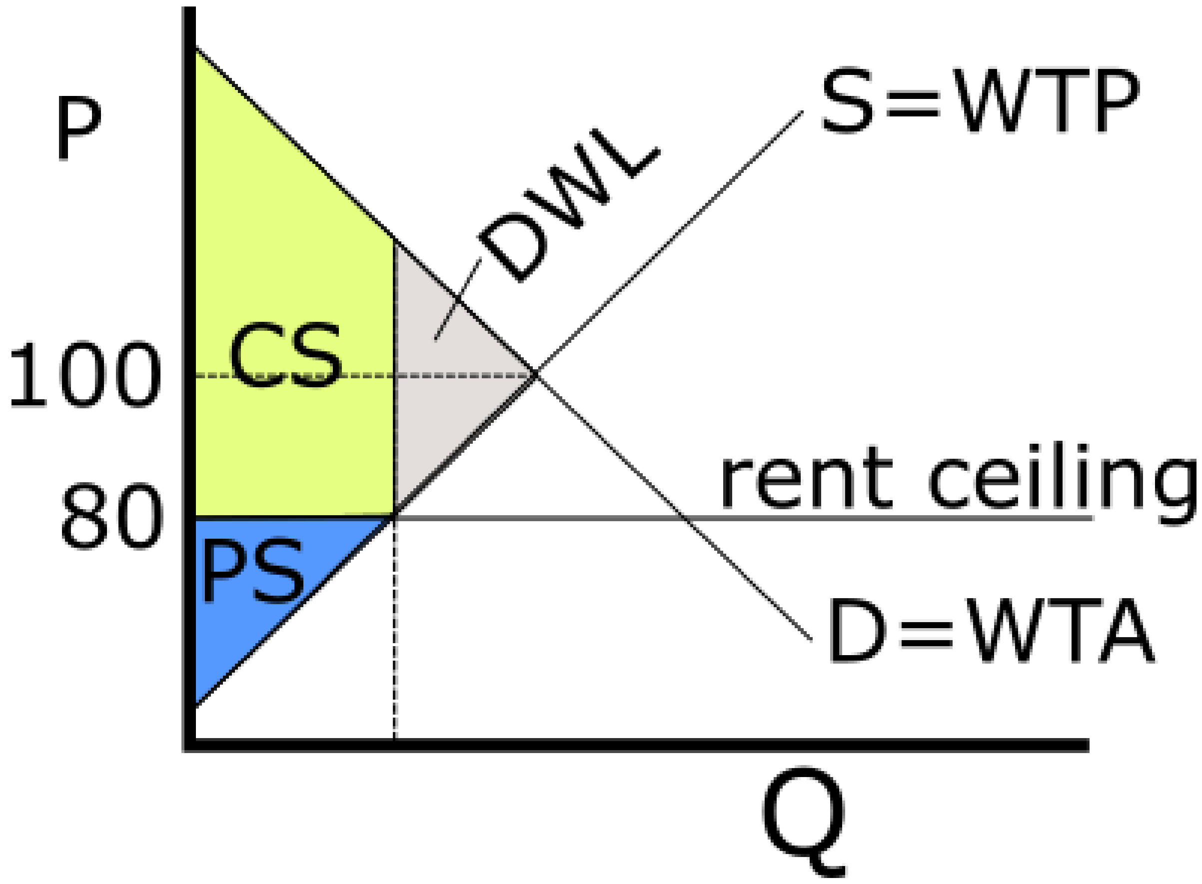
8.12 Rent ceiling

- Some landlords may favour those with a high willingness to pay through, for example, introducing "key" deposits equal to €600.
    - The landlord "claws back" some of the rent (possibly illegally)
    - If this is possible, quantity supplied may increase back to the unregulated equilibrium quantity



8.12 Rent ceilings: welfare effects

- Rent control creates **dead weight loss** (DWL) because at the new, lower, quantity of units that get traded, WTP > WTA at the margin (unexploited gains from trade).
- Rent control reduces producer surplus (PS)
- Rent control may increase consumer surplus (CS)
- But, some of the CS gains may be lost through search costs



- And some consumers no longer get any CS at all because they cannot get housing

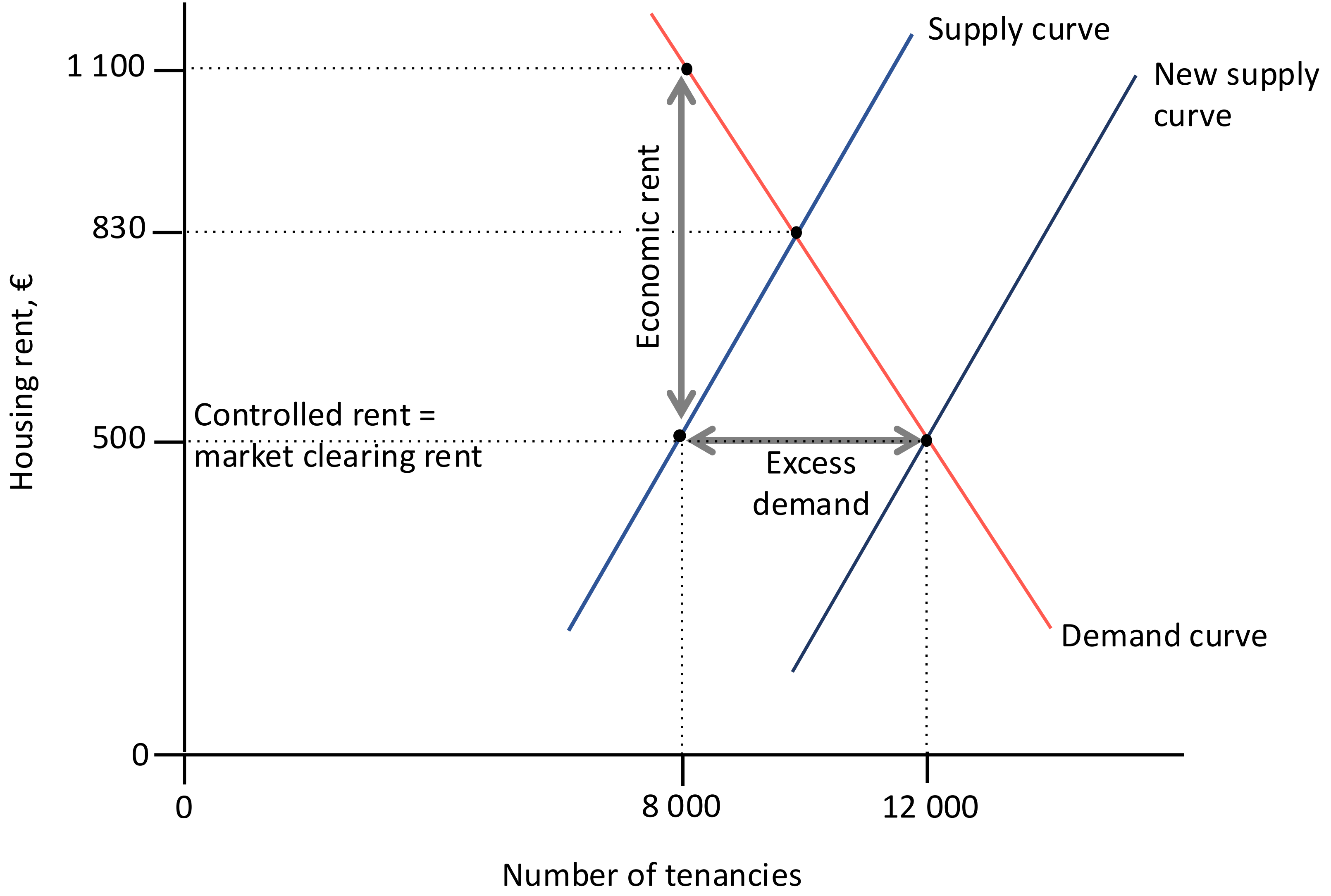
Rent ceiling – efficiency and fairness

- Rent ceilings are inefficient. They cause dead weight loss as well as further losses through unnecessary search costs.
- Are they fair?
  - They transfer surplus from landlords to tenants
  - They benefit those tenants who manage to get a housing unit.
    - Long-time residents. Depending on the context, these can be lower-income or higher-income people or some mixture of both.
  - They harm those tenants who do not manage to get a housing unit due to the shortage.
    - Potential new entrants to an area

8.12 Price ceilings – alternative policies

One solution to the problem of shortages is to subsidize production or otherwise encourage supply to increase.

- Subsidies make the cost of the policies clear, while the cost of search and dead weight loss is not easy to see.
- If government policy is successful at increasing supply, the price controls would in theory not be necessary.



Price controls on food

https://www.theguardian.com/global-development-professionals-network/2015/apr/16/venezuela-economy-black-market-milk-and-toilet-paper

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Price controls and scarcity force Venezuelans to turn to the black market for milk and toilet paper

Thu 16 Apr 2015

In Petare, a giant slum overlooking Caracas from the east, hustlers known as *buhoneros* sell their goods at a busy intersection. "I've got milk, toilet paper, coffee, soap…" said 30-year-old Carmen Rodríguez, pointing to her wares by the side of a road busy with honking motorbikes, cars and buses. "Of course they cost more than the government say they should. We have to queue up to get them or buy them from someone who has done. We're helping people get the basics."

…

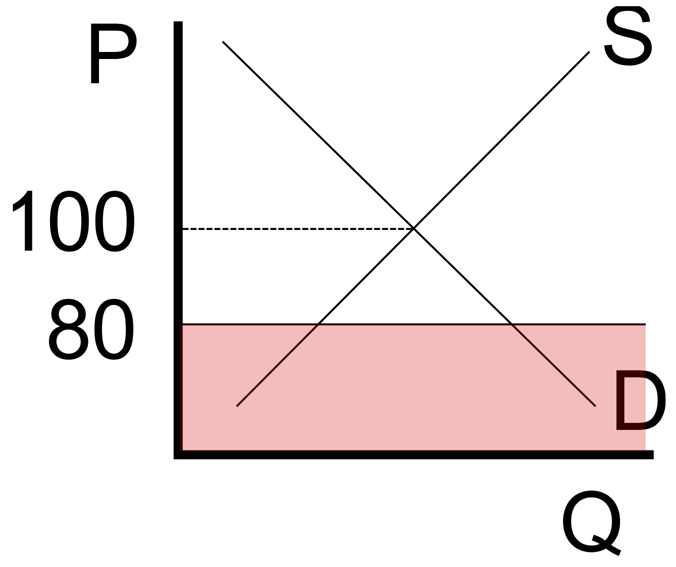
The scenes at Petare's intersection, 23 de enero's streets and Catia's supermarkets are manifestations of an economy in tatters: one in which people buy milk, toilet paper and shampoo at inflated prices because supermarkets, with long queues outside, are near empty…

People queue at a supermarket for limited stock of goods available at government-set prices

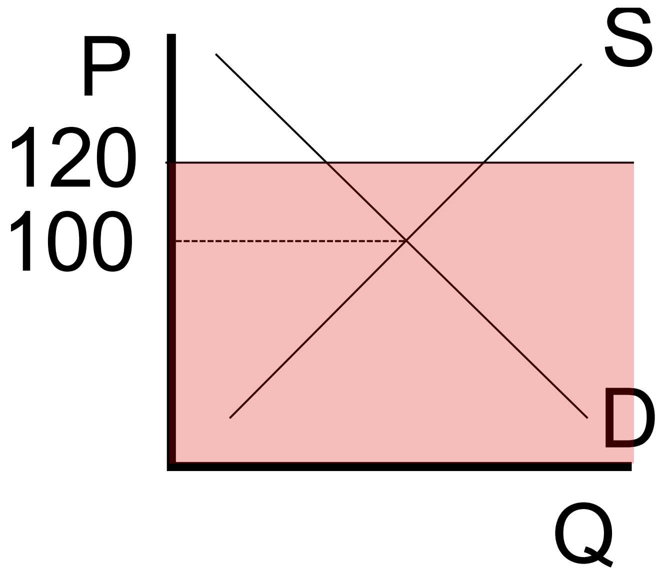
Binding and non-binding price floors

- A price floor is a legally mandated **minimum** price
- If the equilibrium price with no government intervention is R100, then a price floor of R80 would be **non-binding** and have no effect.

 (the shaded area shows illegal prices)



- If the equilibrium price with no government intervention would be R100, then a price floor of R120 would be **binding**.



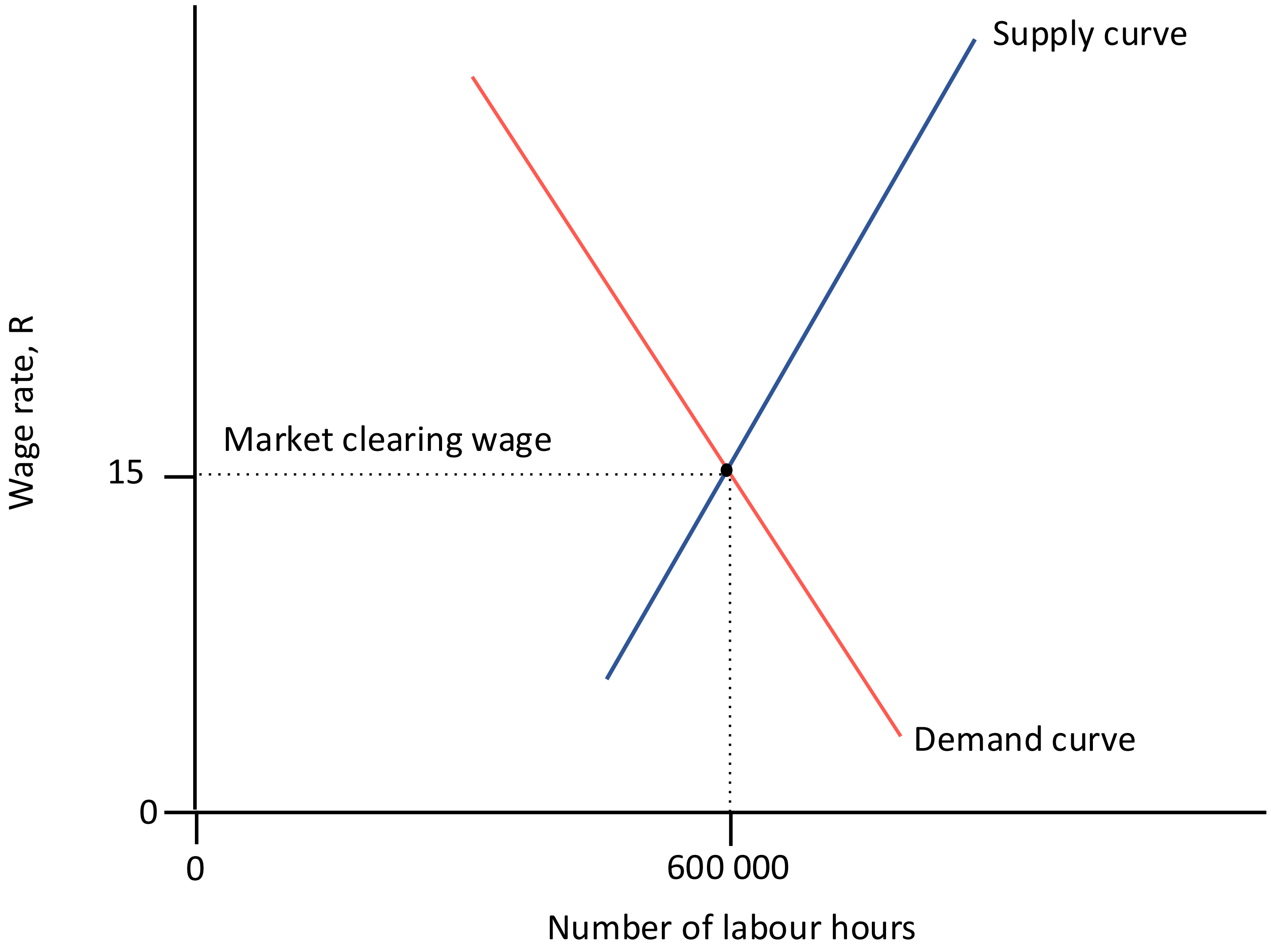
Price floors: Minimum wages

We will assume a perfectly competitive labour market

- The price of labour is the wage rate
- The demand for labour is downward-sloping:
  - Firms demand hours of labour because they need labour as an input to produce goods and services, which they sell to make profit.
  - Firms are motivated by profit. They will employ workers if it is profitable to do so.
  - Firms face declining marginal productivity of labour. At higher numbers of work-hours hired, the revenue generated by an hour of labour declines.
  - Consequently, the wage they can pay (without running a loss) also declines as production increases.
  - Therefore: At higher wage rates, firms will hire fewer work-hours
- The supply of labour is upward-sloping:
  - Households/individuals supply hours of labour. As the wage rate increases, more people want to work.

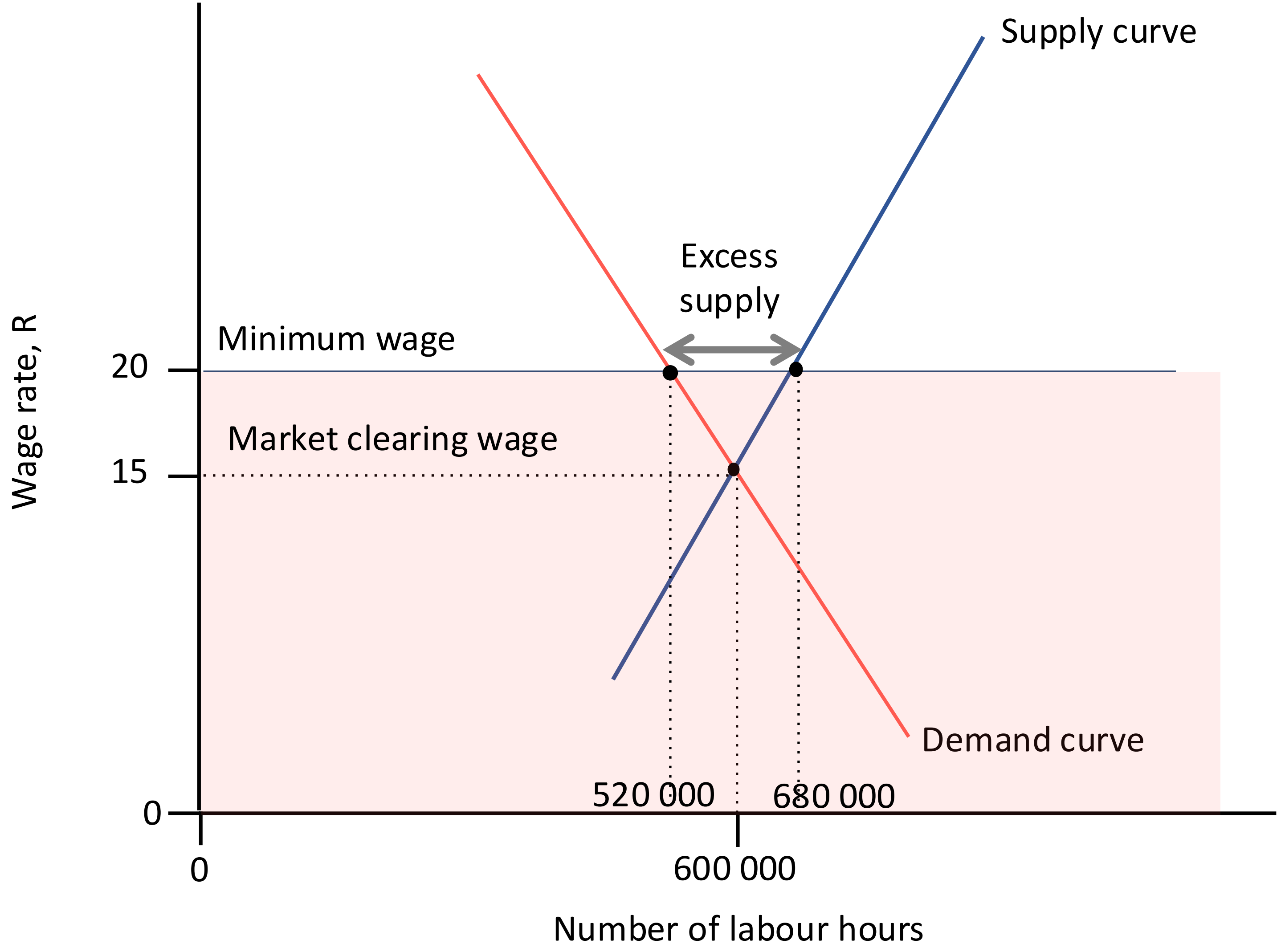
Price floors: Minimum wages

The labour market is initially in equilibrium at a wage rate of R15 an hour, and the equilibrium quantity of labour hours is
600 000



Minimum wages

- Government decides to introduce a minimum wage of R20 per hour
  - It is *illegal* to pay a wage that is *lower* than the minimum wage set (shaded area)
  - The market clearing wage of R15 per hour is illegal
- There is now a surplus (excess supply)of labour hours in the labour market
  - Qd = 520 000 at R20
  - Qs = 680 000 at R20
  - Surplus of 160 000 labour hours



Minimum wages

- Minimum wages can lead to involuntary unemployment
- Similar to price ceilings, binding price floors create dead weight loss and search costs
- They transfer surplus from employers to workers (those who manage to get a job)
- They harm those workers who remain unemployed